

SMOHIT ARMAN SWAIN

192572092

CSA0802 PYTHON PROGRAMMING

ANALYICAL PROBLEM

1.

The screenshot shows the SIMATS Online Programming Lab interface. At the top, there's a header with the SIMATS Engineering logo, the text "SIMATS Online Programming Lab", and the user's name "Smohit Arman Swain" with ID "192572092RC". Below the header, the page is titled "Analytical Problem Solving2". On the left, there's a "QUESTIONS" sidebar with five items: Q1 Harshad number (10 marks), Q2 GCD (10 marks), Q3 sum to a target value (10 marks), Q4 Bitwise Operators (10 marks), and Q5 binary search (10 marks). The main area displays the selected question "Q1 Harshad number" (10 marks). The question text states: "A number is called a Harshad number (or Niven number) if it is divisible by the sum of its digits. Example: 18 → digit sum = 9, 18 % 9 == 0. Write a Python program to read a number N from the user and check whether it is a Harshad number. Also print all Harshad numbers between 1 and 500." Below the question, there's a "Your Answer" section with a text area containing a Python function:

```
def digit_sum(num):  
    s=0  
    while num>0:  
        s+=num%10  
        num//=10  
    return s
```

 and a "Save Answer" button. At the bottom left of the sidebar, it says "5/5 answered".

2.

The screenshot shows a debugging task. The text reads: "The following Python program is intended to find the GCD (Greatest Common Divisor) of two numbers using the Euclidean algorithm. It contains FOUR errors. Identify each error and write the corrected line." Below this, the code is shown:

```
a = int(input("Enter first: ")) b = int(input("Enter second: "))  
while b = 0: temp = a a = b b = temp % a  
print("GCD is", a)
```

 A yellow box with a bug icon and the text "Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below." is present. Below this, there's a "Your Fixed Code" section with a code editor containing the corrected code:

```
1 a=int(input("Enter first: "))  
2 b=int(input("Enter second: "))  
3  
4 while b !=0:  
5     temp = a  
6     a = b  
7     b = temp % b  
8 print("GCD is",a)
```

 To the right of the code editor, there's a "Test Input" section with a text area containing "24" and "18". Below that, there's an "Output" section with a text area containing "Enter first: Enter second: GCD is 6".

3.

sum to a target value

Complete the missing lines in the Python program below. The program finds all pairs of elements in a list that sum to a target value. Fill in all seven blanks.

```
def find_pairs(lst, target):
    seen = ____ pairs = []
    for num in ____:
        complement = target - ____
        if complement in ____:
            pairs.append((complement, num))
            _____.add(num)
    return ____
data = [int(x) for x in input("Enter numbers: ").split()]
t = int(input("Target: "))
result = ____ (data, t)
print("Pairs:", result)
```

Your Answer

```
data = [int(x) for x in input("Enter number: ").split()]
t = int(input("Target: "))

return = find_pairs(data, t)
print("pairs:", result)
```

 Save Answer

4.

Q4 Bitwise Operators

10 marks

Trace the following Python program step by step. Write the exact output. Also state the data type of each variable after assignment.

```
a = 0b1101
b = 0o25
c = 0x1A
result = (a & b) | c
print(f"a={a}, b={b}, c={c}, result={result}")
print(bin(result), oct(result), hex(result))
print(result >> 2, result << 1, ~result)
```

Your Answer

```
result = (a&b)|c
bitwise and

a=13=01101
b=21=10101
```

 Save Answer

The following Python program tries to perform binary search on a rotated sorted array. It has FOUR errors. Identify each error with the line number and rewrite the correct program.

```
def rotated_binary_search(arr, target):
    low = 0
    high = len(arr) - 1
    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == target:
            return mid
        if arr[low] <= arr[mid]:
            if target >= arr[low] and target < arr[mid]:
                high = mid - 1
            else:
                low = mid + 1
        else:
            if target > arr[mid] and target <= arr[high]:
                low = mid + 1
            else: high = mid - 1
    return -1
data = [4,5,6,7,0,1,2]
t = input("Target: ")
result = rotated_binary_search(data, t)
print("Found at:", result)
```

5.

```
1 def rotated_binary_search(arr, target):
2     low = 0
3     high = len(arr) - 1
4
5     while low <= high:
6         mid = (low + high) // 2
7
8         if arr[mid] == target:
9             return mid
10
11        if arr[low] <= arr[mid]:
12            if target >= arr[low] and target < arr[mid]:
13                high = mid - 1
14            else:
15                low = mid + 1
16        else:
17            if target > arr[mid] and target <= arr[high]:
18                low = mid + 1
19            else:
20                high = mid - 1
21    return -1
22 data = [4,5,6,7,0,1,2]
23 t = int(input("Target: "))
24
25 result = rotated_binary_search(data, t)
26 print("Found at:", result)
```

0

Output

Target: Found at: 4

 Pending manual review